

MERIDIAN NAS, MS, USA

WMO: 722345

Lat: 32.550N

Lon: 88.567W

Elev: 271

StdP: 14.55

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03866

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	26.1	7.8	8.3	30.7	12.5	10.5	33.2	21.1	44.2	19.0	43.4	3.8	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	96.1	75.7	94.2	75.8	92.4	75.7	79.7	90.0	78.5	88.8	77.8	87.9	6.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	141.3	84.5	75.7	135.6	83.2	74.9	132.3	82.5	43.2	90.1	42.2	89.4	41.2	88.1	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.2	14.5	12.1	DB	15.3	98.7	4.6	2.4	12.1	100.4	9.4	101.8	6.8	103.1	3.5	104.8
			WB	13.8	81.9	4.5	2.5	10.6	83.7	8.0	85.2	5.5	86.6	2.2	88.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	45.7	50.5	57.6	63.9	72.5	79.3	81.2	80.8	76.3	65.5	53.9	49.3
	DBStd	14.78	10.48	10.76	9.69	7.31	5.90	4.14	3.25	3.56	5.37	8.00	8.57	9.98
	HDD50	570	212	116	42	3	0	0	0	0	0	2	55	140
	HDD65	2343	602	420	263	107	15	0	0	0	2	94	345	495
	CDD50	5967	78	129	278	420	696	879	968	956	788	483	174	118
	CDD65	2266	3	12	35	74	246	429	503	491	341	111	13	8
	CDH74	21763	7	68	290	730	2268	4379	4977	4784	3133	1027	81	19
	CDH80	9595	0	6	46	147	862	2106	2408	2304	1392	314	10	0
(14) Wind	WSAvg	4.5	5.4	5.6	5.7	5.3	4.1	3.7	3.6	3.4	3.6	4.2	4.5	5.0
(15) Precipitation	PrecAvg	56.3	5.6	5.2	6.2	6.0	4.5	3.9	5.1	3.7	3.0	3.1	4.1	5.6
	PrecMax	78.8	12.1	11.3	15.0	15.2	13.5	12.4	11.4	8.6	10.2	9.8	11.3	12.1
	PrecMin	36.4	1.0	1.4	1.8	0.6	0.5	0.2	1.0	0.4	0.2	0.0	0.7	1.1
	PrecStd	12.2	2.7	2.8	3.3	4.0	3.2	2.6	2.8	2.0	2.2	2.8	2.6	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.8	80.6	84.2	86.1	93.0	97.9	97.3	98.9	96.2	90.3	81.1	76.5
		MCWB	62.9	67.2	65.4	68.3	73.9	74.6	75.8	76.2	73.6	71.0	67.8	67.8
	2%	DB	70.6	75.7	81.0	83.9	90.2	95.2	95.1	95.9	93.4	87.0	75.9	72.6
		MCWB	62.6	64.2	64.8	67.6	72.7	75.3	76.8	76.1	73.6	70.3	65.1	65.8
	5%	DB	66.4	71.7	77.3	81.7	87.8	92.8	93.3	93.4	91.0	83.4	72.7	69.3
		MCWB	59.6	62.5	63.0	66.3	71.6	75.2	76.8	76.6	74.0	68.9	63.0	63.6
	10%	DB	63.0	68.0	73.5	78.8	85.2	90.5	91.2	90.9	88.4	80.1	69.3	65.7
		MCWB	57.7	60.2	61.2	64.9	70.6	75.1	76.5	76.4	73.5	67.7	61.6	60.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.2	69.6	69.8	72.9	77.6	80.0	81.9	80.7	78.7	76.6	71.7	70.2
		MCDB	71.4	76.0	76.3	81.0	87.2	90.3	90.6	90.6	88.5	82.8	76.5	74.2
	2%	WB	64.1	67.5	67.5	70.9	75.4	78.4	79.8	79.5	77.4	74.6	68.6	67.2
		MCDB	68.1	73.4	75.1	79.1	85.0	89.1	89.7	90.3	87.1	82.1	72.5	71.2
	5%	WB	61.8	65.0	65.7	69.2	74.1	77.5	78.6	78.4	76.3	72.7	66.0	64.6
		MCDB	66.0	70.4	73.9	77.3	83.4	88.0	88.7	88.8	85.3	79.3	70.2	68.2
	10%	WB	58.6	61.6	63.9	67.6	72.8	76.5	77.7	77.5	75.2	70.7	63.0	61.8
		MCDB	62.6	67.3	71.2	75.7	81.6	86.7	87.9	87.5	83.6	76.5	68.0	64.8
(35) Mean Daily Temperature Range	5% DB	MDBR	20.6	22.0	23.4	24.2	22.2	20.9	19.4	19.9	21.7	23.5	23.4	19.7
		MCDBR	22.9	24.4	25.4	25.9	24.7	23.5	21.8	23.1	24.5	25.6	25.0	21.4
		MCWBR	16.0	14.4	12.6	11.9	9.8	8.2	7.6	7.4	8.6	11.1	15.2	15.0
	5% WB	MCDBR	18.6	20.2	21.0	20.9	20.5	20.4	19.7	19.8	19.5	19.4	19.9	17.6
		MCWBR	15.1	13.9	12.4	11.5	9.4	8.5	8.1	7.5	7.8	10.2	15.0	13.7
(40) Clear-Sky Solar Irradiance	taub		0.326	0.329	0.362	0.396	0.429	0.472	0.516	0.513	0.434	0.371	0.343	0.323
	taud		2.517	2.510	2.431	2.331	2.279	2.190	2.093	2.098	2.304	2.456	2.480	2.538
	Ebn at Noon		284	294	291	284	274	261	248	247	264	276	275	279
	Edh at Noon		27	30	35	40	43	47	51	50	39	31	28	25
(44) All-Sky Solar Radiation	RadAvg		863	1034	1392	1750	1927	1994	1904	1768	1582	1319	978	744
	RadStd		74	143	122	139	129	150	108	123	119	169	98	74

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+238	+157

Nomenclature: See separate page